School of Information and Physical Sciences
SENG4430 – Software Quality

Assessment 1: Group Report (30%)
Due Date: 11:59 pm on 24 Apr (Week 12)

Design Problem

The following is the design problem that each group will undertake:

Computer systems and the software that supports them are embedded in almost every aspect of modern life. As a result, expectations have shifted from simple access to dependable performance, which encompasses security, safety, availability, maintainability, and high-quality user experiences. In parallel, software development has evolved toward agile norms, with lifecycles designed to accommodate continual requirements change, interdisciplinary and distributed teams, asynchronous workflows, and frequent updates.

Maintaining software quality in this environment is difficult. Nevertheless, effective tool support can substantially reduce the risks associated with poor delivery and weak quality management by automating routine activities and generating evidence to guide decisions, particularly those related to prioritisation and resource allocation.

You are part of a small software engineering start-up, and your company has been approached by an **industry organisation** to develop a large software quality support tool. To win the contract, the client would like you to demonstrate your expertise in software engineering and software quality assurance. You will develop a **scalable software quality tool prototype** and produce formal **design documentation** and **quality case** for this prototype.

The scope and aspects of software quality to be considered are open but must, at a minimum, include **usability, reliability, security,** and **maintainability**. The client has also specified that the tool will need to be in **Java**, that you will need to identify your own datasets to test your tool, and that the **tool can evaluate itself**, i.e., the software tool you develop must be able to test itself for software quality.

Project Organisation

You are expected to assemble a **team of 5 to 6 software developers** to work on this project. **Each developer** will take ownership of **one software quality aspect** (excluding usability). For **each software quality aspect**, each developer will develop **one software quality metric**. The **usability** software quality aspect should be **owned by the group**.

You will use an agile software development methodology and work towards weekly sprints. Sprints will be reviewed across the project by the client. Both the prototype code and documentation will be developed and maintained in an agile fashion. Teams are responsible for producing a meeting agenda for each sprint meeting and taking minutes to document progress and actions.

The client's company uses IEEE standards and requires that the IEEE Std 1016-2009 (IEEE Standard for Information Technology – Systems Design – Software Design Descriptions) being used as the basis for the system design document. The client is also interested in how your company can work with standardised templates. Thus, they have provided teams with an example Test Plan template from an Enterprise Project Management Methodology (EPMM) for use on this project.

Deliverables

In this assessment, you are required to prepare a **working prototype** and the following documentations: **system design document, test plan, quality case document, evidence of project management,** and **peer evaluation form**.

Working Prototype

The software quality support tool should be written in Java and at least cover quality characteristics such as usability, reliability, security, and maintainability. It must also be capable of self-testing.

System Design Document

The system design document should conform to the IEEE Std 1016-2009. Detail your design process, algorithm, rational, observations, testing method.

Test Plan

Using the provided EPMM test plan template, document the testing process throughout the entire project development.

Quality Case Document

The quality case document should include example test sets of software used in testing the software quality tool. Evidence of successful quality tests as well as verification and validation should be provided.

Evidence of Project Management

The submission of minutes of meeting is compulsory. Each minutes of meeting should contain attendance, action lists, tasks completed, major project decisions and a commentary on group dynamics. Additional evidence should also be included, but not limited to project action plan, Gantt charts, Kanban, etc.

Peer Evaluation Form

Every team member should submit a peer evaluation form to fairly judge the team dynamics. In case all members have a general consensus on equal contribution, a peer evaluation form may be omitted.

You should hand in your code as well as all documentations to Canvas by **11:59 pm on 24 Apr (Week 12)**. All documentations (except peer evaluation form) should be submitted in PDF format and one copy per group is sufficient.

The marking rubric for this assignment is available with this assignment specification on Canvas. Note that there is no target page length for all documentations. Quality, criteria coverage, and team effort are better than quantity. Be sure that you address each of the assessment criteria.

Late Submission and Academic Integrity

1. Assignments submitted after the deadline will be penalized by 10% per day late (including weekend). For example, an assignment worth 20% marked at 78% may be graded as:
   - On time: 0.78 * 1.00 * 20 = final mark = 15.6
   - 1 day late: 0.78 * 0.90 * 20 = final mark = 14.04
   - 3 days late: 0.78 * 0.70 * 20 = final mark = 10.92
   - 5 days late: 0.78 * 0.50 * 20 = final mark = 7.8

2. As per the Student Conduct Rule, any work submitted for assessment must be your own original work, and as such, the assignment report will be checked for plagiarism. A plagiarized assignment will receive a zero score and be reported to SACO.
3. GenAI usage: GenAI tools such as Microsoft Copilot, ChatGPT, and other similar tools **cannot** be used in the writing or drafting of any work and diagrams submitted for the assessment. The use of GenAI tools is **only acceptable** as an aid in helping you understand the assessment requirements, but be wary about errors and hallucinations.

Mark Allocation

Refer to the complete marking rubrics (SENG4430-Assessment-01-Grading-Rubrics-2026-S1.xlsx) on Canvas.